

Answer to Practice Problems of Exam 1.

1. (a) $\int_0^{\pi/2} \frac{\cos \theta}{1+\sin \theta} d\theta$

let $u = \sin \theta$, then $du = \cos \theta d\theta$

$\theta = 0 \rightarrow u = 0$

$\theta = \frac{\pi}{2} \rightarrow u = 1$

$$\int_0^{\pi/2} \frac{\cos \theta}{1+\sin \theta} d\theta = \int_0^1 \frac{1}{1+u} du$$

$$= \ln(1+u) \Big|_{u=0}^{u=1}$$

$$= \ln 2 - \ln 1$$

$$= \ln 2.$$

(b) let $u = \ln x$, $dv = \frac{1}{x^2} dx$

$$\Rightarrow du = \frac{1}{x} dx, v = -\frac{1}{x}$$

thus

$$\int \frac{\ln x}{x^2} dx = -\frac{1}{x} \ln x - \int -\frac{1}{x} \cdot \frac{1}{x} dx$$

$$= -\frac{\ln x}{x} + \int \frac{1}{x^2} dx$$

$$= -\frac{\ln x}{x} - \frac{1}{x} + C$$

$$= -\frac{1}{x} (\ln x + 1) + C$$

(c) let $u = y$, $dv = e^{-2y} dy$

$$\Rightarrow du = dy, v = -\frac{1}{2} e^{-2y}$$

Thus

$$\int y e^{-2y} dy = -\frac{y}{2} e^{-2y} - \int (-\frac{1}{2} e^{-2y}) dy$$

$$= -\frac{1}{2} y e^{-2y} - \frac{1}{4} e^{-2y} + C$$

$$= -\frac{1}{4} e^{-2y} (2y + 1) + C$$

(d) let $x = \sqrt{t}$, then $t = x^2$, $dt = 2x dx$

$$\int \cos \sqrt{t} dt = \int (\cos x) (2x dx)$$

$$= 2 \int x \cos x dx \quad \text{integration by parts.}$$

$$= 2(x \sin x - \int \sin x dx)$$

$$= 2(x \sin x + \cos x) + C = 2(\sqrt{t} \sin \sqrt{t} + \cos \sqrt{t}) + C$$

2. (a) $\int \sin^2 \theta \cos^2 \theta d\theta = \int \sin^2 \theta \cos^2 \theta \sin \theta d\theta$

$$= \int (1 - \cos^2 \theta) \cos^2 \theta \sin \theta d\theta$$

$$\xrightarrow{u = \cos \theta} \int -(1 - u^2) u^2 du.$$

$$du = -\sin \theta d\theta$$

$$= \int u^4 - u^2 du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{\cos^5 \theta}{5} - \frac{\cos^3 \theta}{3} + C$$

(b) $\int \tan^5 \theta \sec^2 \theta d\theta.$

$$= \int \tan^4 \theta \sec^2 \theta \tan \theta \sec \theta d\theta$$

$$\xrightarrow{u = \sec \theta} \int (u^2 - 1)^2 u^2 du$$

$$du = \tan \theta \sec \theta d\theta$$

$$= \int u^6 - 2u^4 + u^2 du$$

$$= \frac{u^7}{7} - \frac{2}{5} u^5 + \frac{u^3}{3} + C$$

$$= \frac{\sec^7 \theta}{7} - \frac{2}{5} \sec^5 \theta + \frac{1}{3} \sec^3 \theta + C$$

(c) $\int \cos^2 \theta d\theta$

$$= \int \frac{1}{2} (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2} (\theta + \frac{1}{2} \sin 2\theta) + C$$

$$= \frac{\theta}{2} + \frac{1}{4} \sin 2\theta + C$$

i.e. $x = \cos^{-1}(\frac{1}{x})$

(d) let $x = \sec \theta$, then $x = \sec^{-1} \theta.$

$$dx = \tan \theta \sec \theta d\theta.$$

$$x = 1 \rightarrow \theta = \cos^{-1}(\frac{1}{1}) = (\cos^{-1} 1) = 0$$

$$x = 2 \rightarrow \theta = \cos^{-1}(\frac{1}{2}) = \frac{\pi}{3}$$

$$\Rightarrow \int_1^2 \frac{\sqrt{x^2 - 1}}{x} dx = \int_0^{\pi/3} \frac{\tan \theta}{\sec \theta} \cdot \tan \theta \sec \theta d\theta$$

$$= \int_0^{\pi/3} \tan^2 \theta d\theta$$

$$= \int_0^{\pi/3} (1 + \tan^2 \theta) - 1 d\theta$$

$$= [\tan \theta - \theta]_{\theta=0}^{\theta=\pi/3}$$

$$= (\tan \frac{\pi}{3} - \frac{\pi}{3}) - 0$$

$$= \sqrt{3} - \frac{\pi}{3}$$

$$(e) x^2 + 3x - 10 = (x+5)(x-2)$$

$$\Rightarrow \frac{x-9}{x^2+3x-10} = \frac{A}{x+5} + \frac{B}{x-2}$$

$$= \frac{A(x-2) + B(x+5)}{(x+5)(x-2)}$$

$$= \frac{(A+B)x + (-2A+5B)}{(x+5)(x-2)}$$

$$\Rightarrow \begin{cases} A+B=1 \\ -2A+5B=-9 \end{cases} \Rightarrow \begin{cases} A=2 \\ B=-1 \end{cases}$$

Thus

$$\int \frac{x-9}{x^2+3x-10} dx = \int \frac{2}{x+5} - \frac{1}{x-2} dx$$

$$= 2 \ln|x+5| - \ln|x-2| + C$$

$$3. (a). \int \frac{dx}{x \ln x} \quad \begin{matrix} u = \ln x \\ du = \frac{1}{x} dx \end{matrix} \int \frac{1}{u} du$$

$$= \ln|u| + C$$

$$= \ln|\ln x| + C$$

Thus

$$\int_2^{+\infty} \frac{dx}{x \ln x} = \lim_{t \rightarrow +\infty} \int_2^t \frac{dx}{x \ln x}$$

$$= \lim_{t \rightarrow +\infty} [\ln|\ln x|]_{x=2}^{x=t}$$

$$= \lim_{t \rightarrow +\infty} \ln|\ln t| - \ln|\ln 2|$$

$$= +\infty$$

→ divergent

$$(b) \frac{1}{\sqrt{6-y}} \text{ is discontinuous at } y=6$$

$$\int_2^6 \frac{dy}{\sqrt{6-y}} = \lim_{t \rightarrow 6^-} \int_2^t \frac{dy}{\sqrt{6-y}}$$

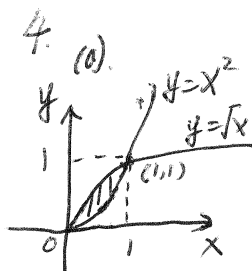
$$= \lim_{t \rightarrow 6^-} [-2\sqrt{6-y}]_{y=2}^{y=t}$$

$$= \lim_{t \rightarrow 6^-} (-2\sqrt{6-t}) - (-2\sqrt{4})$$

$$= \lim_{t \rightarrow 6^-} (-2\sqrt{6-t} + 4)$$

$$= 0 + 4$$

$$= 4 \rightarrow \text{Convergent}$$



$$\begin{cases} y = \sqrt{x} \\ y = x^2 \end{cases} \Rightarrow \sqrt{x} = x^2$$

$$x^2 - \sqrt{x} = 0$$

$$\sqrt{x}(x^{3/2} - 1) = 0$$

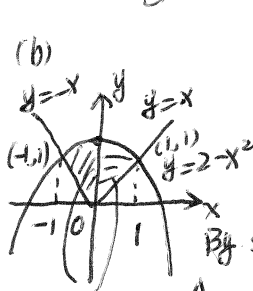
$$x=0 \text{ or } x=1$$

$$A = \int_0^1 \sqrt{x} - x^2 dx$$

$$= \left(\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right)_{x=0}^{x=1}$$

$$= \left(\frac{2}{3} - \frac{1}{3} \right) - (0)$$

$$= \frac{1}{3}$$



$$\begin{cases} y = x \\ y = 2-x^2 \end{cases} \Rightarrow x = 2-x^2$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2, x = 1$$

remove since $x \geq 0$.

By symmetry, we have

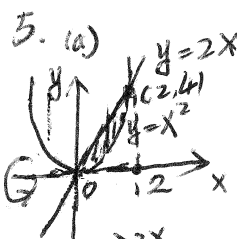
$$A = 2 \int_0^1 (2-x^2) - x dx$$

$$= 2 \left[2x - \frac{x^3}{3} - \frac{x^2}{2} \right]_{x=0}^1$$

$$= 2 \left[2 - \frac{1}{3} - \frac{1}{2} \right] - (0)$$

$$= 2 \cdot \frac{7}{6}$$

$$= \frac{7}{3}$$



$$\begin{cases} y = x^2 \\ y = 2x \end{cases} \Rightarrow x^2 = 2x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x=0 \text{ or } x=2$$



$$A(x) = \pi(2x)^2 - \pi(x^2)^2$$

$$= \pi(4x^2 - x^4)$$

$$x^2 \Rightarrow V = \int_0^2 A(x) dx$$

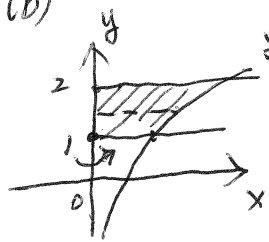
$$= \int_0^2 \pi(4x^2 - x^4) dx$$

$$= \pi \left[\frac{4}{3} x^3 - \frac{1}{5} x^5 \right]_{x=0}^2$$

$$= \pi \left[\frac{4}{3} \cdot 8 - \frac{1}{5} \cdot 32 \right] - (0)$$

$$= \frac{64}{5} \pi$$

(b)



$$y = \ln x \quad (x = e^y)$$

$$A(y) = \pi (e^y)^2$$

$$= \pi e^{2y}$$

$$V = \int_1^2 A(y) dy$$

$$= \int_1^2 \pi e^{2y} dy$$

$$= \pi \left(\frac{e^{2y}}{2} \right)_{y=1}^{y=2}$$

$$= \frac{\pi}{2} (e^4 - e^2)$$

